Reverse- and Reengineering
Homework 1

Winter 2026/27
Date: 20.10.2026

Prof. Dr. Christian Hammer
Dr. Mohammad Rezaalipour

This sheet is not for submission. However, we strongly encourage you to solve the problems and bring your solutions to the tutorials for discussion.

1 Intra-Procedural CFG, PDG, and Program Slicing

Consider the following program:

```
1  int main() {
2    int x, y, prod, lcm, gcd=1;
3    scanf("%d", &x);
4    scanf("%d", &y);
5    for (int i=1;i <= x && i <= y; ++i) {
6      if (x%i==0 && y%i==0) {
7        gcd=i;
8      }
9    }
10   prod=x*y;
11   lcm=prod/gcd;
12   printf("gcd=%d",gcd);
13   printf("prod=%d", prod);
14   printf("lcm=%d", lcm);
15 }
```

1. Draw the intra-procedural CFG for the program.
2. Draw the program dependence graph for the program above. Assume `scanf` defines the variable and `printf` uses the variable.
3. Compute the backward slice for line 13 and line 14.
4. Compute the forward slice for the statement `int i=1` at line 5.

2 Inter-Procedural CFG

Draw the inter-procedural CFG for the program below.

```
1  #include <iostream>
2
3  int add(int x, int y) {
4    return x+y;
5  }
6
7  int sub(int x, int y) {
8    return x-y;
9  }
```

```
10 int main()
11 {
12   int x, y, a, b;
13   int (*func)(int, int) = nullptr;
14   std::cin >> x >> a >> b;
15   if (x%2 == 0) {
16     func = &add;
17   } else {
18     func = &sub;
19   }
20   y = func(a,b);
21   std::cout << y;
22 }
```